

Transformers for Suicidal Ideation Detection in Reddit Posts: Evaluating BERT, RoBERTa, and BERTweet

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Abstract

Suicidal ideation is a major public health issue, especially among adolescents and young adults. The rise of anonymous platforms like Reddit has created new opportunities for studying mental health-related language, but also significant methodological and ethical challenges. This paper presents a comparative evaluation of transformer-based models for classifying suicide-related Reddit content under a subreddit-derived proxy labeling scheme, including BERT, RoBERTa, and BERTweet. Going beyond aggregate accuracy, I incorporated recent research on intention modeling, interpretability, and bias while examining practical deployment constraints. I benchmarked these models against a TF-IDF + Logistic Regression baseline and conducted analyses of model errors, qualitative failure cases, and broader ethical implications. Using custom fine-tuning and evaluation pipelines for pretrained transformer models, I found that RoBERTa produced the highest recorded suicide-class F1 in the full-dataset comparison, while BERT and BERTweet also achieved strong performance. I additionally explored the limitations of SHAP and attention-based analyses in this setting and conclude with directions including multilingual adaptation, temporal modeling, improved labeling, and human-in-the-loop frameworks.

1 Introduction

Suicide claims more than 720,000 lives annually and was the third leading cause of death among 15–29-year-olds globally in 2021 (WHO, 2025). Seventy-three percent of suicides occur in low- and middle-income countries. Early detection and intervention remain important public-health goals, while stigma, limited access to care, and indirect disclosure can make risk difficult to identify.

Online platforms such as Reddit contain large volumes of user-generated discussion about mental health, distress, and suicide. Language in these settings can be informal, context-dependent, and

difficult to represent with purely lexical methods. Prior work has documented limitations of feature-engineered approaches and emphasized the need for models that better represent context and intention (Ji et al., 2021; Ji, 2022). Transformer-based models such as BERT, RoBERTa, and BERTweet provide one approach for studying whether richer contextual representations improve classification performance.

However, applying transformer models to mental-health content introduces substantial challenges. Interpretability, fairness, label validity, generalization, and the consequences of false predictions must all be considered before moving from benchmark performance toward real-world use. My goal was therefore not only to compare transformer performance, but also to examine what their errors and interpretability experiments reveal—and do not reveal—about their suitability for high-stakes screening contexts.

The motivation behind this project is both technical and personal. I was drawn to this topic due to my own experiences in a high-pressure school environment that led to an increased awareness of mental health challenges among students, including suicide. I studied Computational Cognitive Science for my undergraduate degree, worked at a mental health clinic, and have written and spoken on mental health advocacy. I hope to carry this interest into my career and build solutions that are impactful, respectful, and grounded in real-world needs.

2 Related Work

2.1 Classical and Clinical Detection

Research on suicide risk has drawn on questionnaire instruments, clinician-administered assessments, electronic health records, suicide notes, and online user content. Ji et al. (2021) organize prior computational work across these data sources and

describe challenges including feature engineering, annotation quality, generalization, and interpretability. Social-media data introduces additional opportunities for scalable text analysis, but its labels and linguistic context differ substantially from clinically collected data.

2.2 Lexical & Feature-Based Machine Learning

Early machine-learning approaches commonly relied on TF-IDF, LIWC, N-gram, syntactic, topic, and other engineered features combined with classifiers such as Logistic Regression and SVMs. Ji et al. (2021) note that these approaches can require substantial feature engineering and domain expertise. Coppersmith et al. (2014), for example, demonstrated that linguistic and behavioral features, including LIWC and word- and character-level language models, could distinguish self-reported mental-health cohorts on Twitter.

2.3 Deep Learning & Hybrid Approaches

The adoption of CNNs, RNNs, and attention-based models enabled learned representations to replace some manual feature engineering. Yates et al. (2017) applied convolutional neural networks to depression and self-harm risk assessment in online forums and demonstrated improved performance over several prior approaches. These developments helped motivate subsequent work using contextual transformer representations.

2.4 Transformers in Suicidal Ideation Detection

Recent work has increasingly incorporated pre-trained transformer architectures. BERT (Devlin et al., 2018) uses bidirectional contextual representations and was pretrained on BooksCorpus and Wikipedia. RoBERTa (Liu et al., 2019) modifies the BERT pretraining procedure through changes including dynamic masking, substantially more pretraining data, and larger training batches. BERTweet (Nguyen et al., 2020) was pretrained on 850 million English tweets, making it particularly relevant for comparison on informal social-media language.

Benchmark performance, however, is not equivalent to intention understanding. Ji (2022) argues that improved classification scores do not by themselves establish genuine understanding of suicidal intention. This distinction is especially important

when labels are proxies derived from online communities rather than clinical assessments.

2.5 Model Explainability and Risk Detection

I also reviewed approaches to model interpretation, including SHAP (Lundberg & Lee, 2017) and LIME (Ribeiro et al., 2016). I considered ClinicalBERT (Alsentzer et al., 2019), a transformer pretrained on clinical notes, but did not use it because its pretraining domain differs substantially from Reddit discourse.

Interpretability remains an important open problem in high-stakes NLP. Ji et al. (2021) note that high-performing deep models can remain difficult to interpret and call for methods capable of relating predictions to meaningful risk factors. In this project, I therefore explored token-level SHAP, self-attention visualization, and qualitative error analysis while treating these techniques as exploratory rather than as definitive explanations of model reasoning.

3 Methodology

All transformer models were implemented using the HuggingFace Transformers library and fine-tuned using manual PyTorch training loops. The classical baseline was built with scikit-learn. For exploratory interpretability analysis, I applied SHAP (SHapley Additive exPlanations; Lundberg & Lee, 2017) and visualized BERT self-attention.

3.1 Dataset Overview

I used the publicly available *Suicide and Depression Detection* Kaggle dataset containing 232,074 Reddit posts with dataset-provided binary labels of “suicide” and “non-suicide” (Komati, 2021). These labels originate from subreddit-based data collection rather than clinical assessment, so they should be interpreted as proxy labels for suicide-related content rather than verified individual suicidal ideation.

The binary dataset is exactly balanced at 116,037 posts per class. For the full-dataset experiments, I used a stratified 80% training and 20% validation split with `random_state=42`, producing a 46,415-post validation set containing 23,207 suicide-labeled and 23,208 non-suicide-labeled posts. The same dataframe, split proportion, stratification strategy, and split seed were used across the recorded full-dataset transformer experiments.

3.2 Preprocessing and Tokenization

For the classical baseline, text was vectorized with scikit-learn’s `TfidfVectorizer` using unigrams and bigrams, English stop-word removal, and a 5,000-feature cap. No lemmatization or hand-written cleaning rules were applied.

For the transformer experiments, text was passed to HuggingFace tokenization pipelines without manual punctuation stripping or text normalization by the preprocessing code. The full-dataset BERT and recorded BERT/RobERTA comparison used a maximum sequence length of 256 tokens. BERTweet used a maximum sequence length of 128 and invoked its tokenizer normalization option. The 10K auxiliary experiments also used 128-token inputs.

3.3 Models Trained

I evaluated a TF-IDF + Logistic Regression baseline followed by BERT (`bert-base-uncased`), RoBERTa (`roberta-base`), and BERTweet (`vinai/bertweet-base`). For additional evaluation, I used a balanced subset of 10,000 posts, with 5,000 examples from each class and a stratified 80/20 split yielding a 2,000-post validation set.

The primary full-dataset transformer training loops used AdamW with a learning rate of $2e-5$, batch size 16, a linear learning-rate schedule with no warmup, and three training epochs. Validation performance was recorded after each epoch, and checkpoints were saved when validation loss improved. For the standalone full-dataset BERT run, validation loss reached its minimum after epoch 1 (0.0781) before increasing to 0.0820 and 0.1095, suggesting increasing validation loss despite continued reductions in training loss.

Auxiliary BERT interpretability and error-analysis notebooks used separate 10K-subset runs and are treated as supporting analyses rather than as identical replications of the canonical three-model subset comparison.

3.4 Interpretability Tools

SHAP was used experimentally to examine token-level attribution outputs for BERT on both high-probability and more ambiguous predictions. Self-attention weights from BERT’s final transformer layer were also visualized to inspect attention patterns. These analyses were applied only to BERT and are descriptive rather than causal explanations of the model’s predictions.

4 Experiments & Results

4.1 Model Development and Iterative Insights

I began with a TF-IDF + Logistic Regression baseline to establish a classical machine-learning benchmark before fine-tuning BERT, RoBERTa, and BERTweet. Across the recorded runs, the transformer models substantially outperformed the TF-IDF baseline. RoBERTa produced the highest full-dataset suicide-class metrics, while BERT and BERTweet also achieved approximately 0.98 suicide-class F1.

These experiments establish differences in recorded predictive performance, but they do not separately test specific linguistic capabilities such as sarcasm recognition, metaphor understanding, slang handling, or sensitivity to implicit language. Those behaviors therefore remain topics for targeted future evaluation rather than conclusions of the present experiments.

4.2 Performance Evaluation

On the full 46,415-post validation set, the recorded BERTweet run achieved suicide-class precision, recall, and F1 of approximately 0.99, 0.98, and 0.98, respectively, and approximately 0.98 F1 for the non-suicide class. These results show strong performance on this Reddit dataset, although they do not independently establish BERTweet’s performance on particular linguistic phenomena such as slang or sarcasm.

In the recorded full-dataset BERT/RobERTA comparison, RoBERTa achieved suicide-class precision of 0.996, recall of 0.992, and F1 of 0.994, while BERT achieved 0.979, 0.980, and 0.980 respectively. The models used the same dataframe and stratified validation split. Given 23,207 suicide-labeled validation examples, the recorded recall values correspond to approximately 178 false negatives for RoBERTa and 457 for BERT, or about 61% fewer false negatives in the RoBERTa run.

This comparison establishes an observed performance difference but does not isolate its cause. BERT and RoBERTa differ in architecture and pre-training procedure, and each configuration was evaluated only once without seed-variance or significance analysis.

The TF-IDF + Logistic Regression baseline achieved 93% accuracy. It reached precision 0.92 and recall 0.94 for the non-suicide class and precision 0.94 and recall 0.92 for the suicide class, with an F1-score of 0.93 for both classes. The

Model	Dataset	Accuracy	Precision (Suicide)	Recall (Suicide)	F1 (Suicide)	F1 (Overall)
TF-IDF + LR	Full (46,415)	93%	0.94	0.92	0.93	0.93
BERT	Full (46,415)	98%	0.979	0.980	0.980	0.98
BERT	Subset (10K, 2K val)	96%	0.941	0.980	0.960	0.96
RoBERTa	Full (46,415)	99%	0.996	0.992	0.994	0.99
RoBERTa	Subset (10K, 2K val)	98%	0.970	0.981	0.976	0.98
BERTweet	Full (46,415)	98%	0.99	0.98	0.98	0.98
BERTweet	Subset (10K, 2K val)	97%	0.972	0.970	0.971	0.97

Table 1: Performance comparison of all models on the full validation split and the 10K balanced subset. Full-dataset transformer rows reflect the recorded three-epoch runs. Subset metrics are derived directly from the confusion matrices in Table 2.

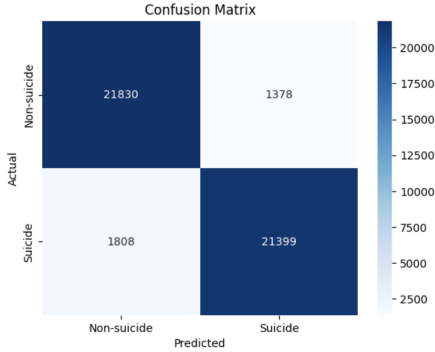


Figure 1: Confusion matrix for the TF-IDF + Logistic Regression model

transformer experiments therefore produced substantially fewer classification errors than the lexical baseline on this dataset.

4.3 Confusion Matrix Analysis

The TF-IDF baseline’s errors were close to symmetric but not identical across classes. Of 23,207 suicide-labeled posts, 1,808 were classified as non-suicide (7.8%). Of 23,208 non-suicide-labeled posts, 1,378 were classified as suicide (5.9%). The baseline therefore produced more false negatives than false positives in this validation split.

4.4 Qualitative Error Analysis

Ji (2022) distinguishes benchmark classification performance from actual understanding of suicidal intention, an important consideration when interpreting high aggregate model scores.

To examine model errors qualitatively, I manually reviewed selected misclassified examples from an auxiliary BERT run on the 10K subset. Among the reviewed false positives were posts describing general distress, disappointment, or suicide in an informational rather than first-person context. One example described a possible suicide the author had heard about while traveling home from school and explicitly framed the post as awareness-raising

rather than personal ideation.

The reviewed false negatives included third-person framing and concern expressed on behalf of another person. In one example, a long post described a friend’s deteriorating mental state and explicitly stated concern that the friend might die by suicide, yet the post carried the dataset’s suicide label and was predicted as non-suicide. These examples suggest hypotheses about possible failure modes, but the manually reviewed sample was too small to establish their population-level frequency. Further examples appear in Appendix A.

4.5 Explainability: SHAP and Attention

I explored SHAP and attention visualization as descriptive tools for examining BERT. The results should be interpreted cautiously because the attribution configuration used here did not yield sufficiently informative token-level explanations to support claims about causal model reasoning.

4.5.1 SHAP on a High-Probability Prediction

In one example, BERT assigned 97.67% probability to the non-suicide class and 2.33% to the suicide class. As shown in Figure 2, the SHAP configuration used in this experiment produced values close to zero across the displayed tokens.

These near-zero values are reported as an empirical outcome of the attribution setup rather than as evidence that no individual token influenced the prediction or that the model relied on any particular internal representation.

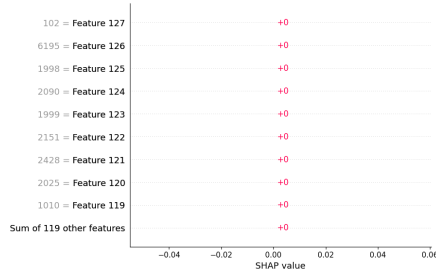


Figure 2: SHAP values produced for a high-probability non-suicide prediction

4.5.2 SHAP on a More Ambiguous Prediction

I also examined examples with model probabilities closer to the decision boundary. In one analyzed case carrying the dataset’s suicide label, BERT assigned approximately 65.5% probability to non-suicide and 34.5% to suicide. The SHAP configuration again produced token-level values very close to zero, and the associated aggregate feature-importance visualization similarly showed little separation among tokens.

The appropriate conclusion is therefore limited: this SHAP configuration did not produce sufficiently informative token-level explanations for either the high-probability or more ambiguous example. The result does not establish whether BERT relied on distributed representations, particular lexical cues, or another internal mechanism.



Figure 3: SHAP token values for an ambiguous BERT prediction

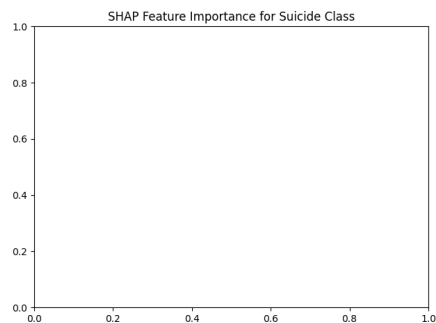


Figure 4: SHAP feature-importance output for the suicide class

4.5.3 Self-Attention Analysis

To complement the SHAP experiment, I visualized the self-attention matrix from BERT’s final transformer layer. Figure 5 shows a strong diagonal pattern as well as several off-diagonal attention relationships. The visualization describes how attention weights were distributed for this particular input, but attention weights alone should not be treated as a causal explanation of the model’s classification decision.

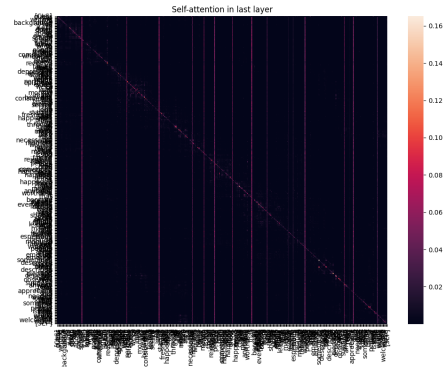


Figure 5: Final-layer BERT attention heatmap for an analyzed input

Taken together, the SHAP and attention experiments were useful primarily for identifying limitations in the interpretability pipeline. They motivate future evaluation with alternative approaches such as text-aware masking, gradient-based attribution, broader samples, and systematic comparison across correctly and incorrectly classified examples.

4.6 Expanded Evaluation on the 10K Subset

I additionally evaluated BERT, RoBERTa, and BERTweet on a balanced 10K sample containing 5,000 examples from each class, with a 2,000-post validation split. The purpose of this experiment was to compare the three architectures under limited fine-tuning and generate directly comparable confusion matrices.

Model	TN	FP	FN	TP
BERT	938	62	20	980
RoBERTa	970	30	19	981
BERTweet	972	28	30	970

Table 2: Confusion matrices for BERT, RoBERTa, and BERTweet on the 2,000-post validation split of the 10K balanced sample.

In this single subset run, RoBERTa produced 19 false negatives, BERT produced 20, and BERTweet produced 30. BERTweet produced the fewest false

positives (28), followed by RoBERTa (30) and BERT (62). Because the differences in recall between RoBERTa and BERT correspond to only one validation example in this run, these values should be interpreted descriptively rather than as a statistically established ranking.

To compare suicide-class metrics directly:

Model	Prec.	Recall	F1
BERT	0.941	0.980	0.960
RoBERTa	0.970	0.981	0.976
BERTweet	0.972	0.970	0.971

Table 3: Suicide-class performance metrics on the 10K subset, derived from Table 2.

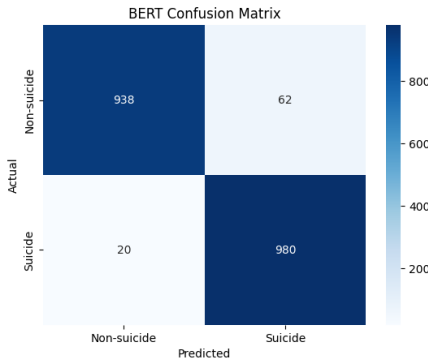


Figure 6: Confusion matrix for BERT

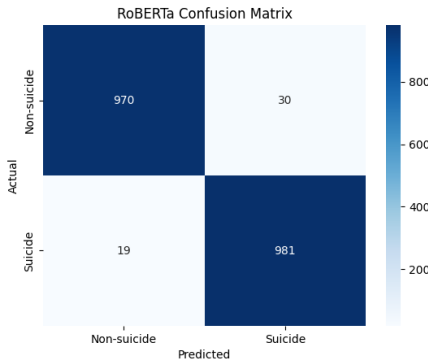


Figure 7: Confusion matrix for RoBERTa

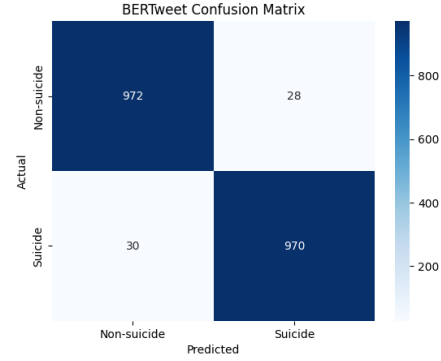


Figure 8: Confusion matrix for BERTweet

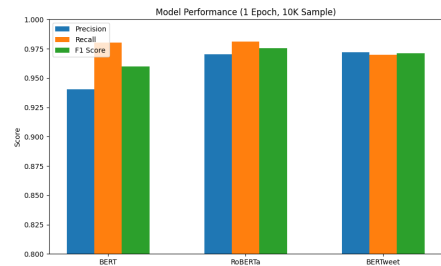


Figure 9: Suicide-class precision, recall, and F1-score comparison across models

The subset experiment provides an additional comparison under limited fine-tuning, while the interpretability experiments demonstrate that high predictive performance does not automatically yield informative explanations of individual predictions.

Future interpretability work could compare correctly and incorrectly classified examples, examine multiple layers and attention heads, use gradient-based attribution methods such as Integrated Gradients, apply text-aware SHAP masking, and aggregate explanations across larger samples.

4.7 Relation to External Benchmarks

Lamichhane (2023) evaluated ChatGPT’s zero-shot performance on a five-class suicidality classification task and reported an F1 score of 0.37 against a majority-class baseline of 0.19. The task differs substantially from the binary proxy-label classification used in this study, so the scores should not be compared as equivalent benchmarks. Instead, the result illustrates the broader distinction between general-purpose zero-shot models and task-specific fine-tuning on labeled domain data.

Technique	Key Observation	Interpretation
SHAP (High Probability)	Token attributions near zero	Current SHAP setup did not yield informative token-level explanations
SHAP (Ambiguous)	Token attributions remained near zero	The same interpretability limitation persisted on a less certain prediction
Attention Map	Strong diagonal pattern with some off-diagonal links	Descriptive attention pattern; not a causal explanation of the prediction
Confusion Matrix	RoBERTa: 19 FN; BERT: 20 FN; BERTweet: 30 FN	RoBERTa had the fewest false negatives in this single subset run
Metric Bar Chart	Suicide F1: RoBERTa 0.976; BERTweet 0.971; BERT 0.960	RoBERTa had the highest recorded suicide-class F1 in this single subset run

Table 4: Summary of interpretability and subset-evaluation observations. SHAP and attention rows describe BERT only.

5 Ethical Considerations and Deployment Constraints

5.1 Bias and Fairness

Model outputs may reflect biases present in training data, and subreddit-derived labels may encode differences in community membership, demographics, language, and posting style in addition to suicide-related content. The dataset used here does not provide the demographic annotation required to measure such effects directly, so fairness across demographic groups was not evaluated.

Future evaluation could incorporate demographic and linguistic subgroup analysis where ethically and legally appropriate, examine annotation bias of the kind discussed by Ji et al. (2021), and measure error disparities across relevant populations. Linthicum et al. (2019) discuss broader ethical and clinical challenges associated with machine-learning-based suicide prediction, including the consequences of prediction errors and difficulties translating algorithmic risk estimates into responsible intervention.

5.2 Interpretability and Clinical Trust

Ensemble architectures offer another direction for balancing predictive performance and transparency. Alghazzawi et al. (2025) combine TF-IDF features with a stacked ensemble including SVM, Logistic Regression, Gradient Boosting, and Decision Tree classifiers, with a Random Forest meta-classifier, while emphasizing both performance and interpretability.

The SHAP analysis in this project did not provide strong local explanations, reinforcing the need to validate interpretability methods themselves before relying on them in high-stakes applications. Future systems could combine multiple explanation methods with structured human review rather

than treating an individual visualization as evidence of model reasoning.

5.3 Real-Time and Longitudinal Use

Mental-health language and online communities change over time, making temporal drift an important concern for deployed systems. Potential strategies include periodic retraining, temporal validation splits, and time-aware modeling approaches such as those explored by Sawhney et al. (2020).

Real-world deployment would also require clear boundaries around user consent, privacy, intervention thresholds, and human oversight. Prediction errors can carry meaningful consequences in either direction, and a classifier trained on subreddit-derived proxy labels should not be treated as a clinical diagnostic system.

Ultimately, automated models should support rather than replace qualified human judgment. Any deployment would require additional validation, ethical oversight, and cross-disciplinary review well beyond the experiments presented here.

6 Future Work and Research Directions

Several directions could extend the current study while addressing its methodological limitations.

6.1 Multilingual and Cross-Cultural Adaptation

The current models were evaluated on English-language Reddit data. Expressions of distress and suicidal ideation differ across languages and cultures, motivating multilingual and cross-cultural evaluation. Zevallos et al. (2024) fine-tune mBERT, XLM-R, and mT5 using a Spanish-language suicide-related tweet corpus translated into five additional languages, demonstrating the feasibility of multilingual modeling while high-

lighting concerns around translation fidelity and cultural nuance.

Future work could combine translated corpora with original non-English social-media data and evaluate performance separately by language rather than assuming that translated training examples preserve culturally specific meaning.

6.2 Temporal and Behavioral Modeling

Most experiments in this study classify posts independently. Future work could incorporate longitudinal patterns using time-aware architectures such as STATENet (Sawhney et al., 2020), model sequences of user activity over time, or select informative utterances from longer conversations using approaches such as Wang et al. (2023).

6.3 Hybrid and Ensemble Methods

Future systems could combine transformer representations with other architectures or engineered features. Lin et al. (2024) explore a RoBERTa-CNN approach and show that data quality materially affects model robustness, while Alghazzawi et al. (2025) demonstrate an explainable stacked ensemble using classical feature-based models.

6.4 Improving Label Quality

The dataset’s labels are derived from subreddit-based collection rather than individual clinical annotation. Community membership therefore functions only as a proxy for the underlying construct of interest.

Future work could incorporate expert or trained-human annotation, explicit uncertainty labels, active learning for ambiguous examples, and evaluation datasets in which individual posts are labeled according to clearly defined risk criteria. Lin et al. (2024) additionally show that manual and OpenAI-assisted data cleaning can materially affect downstream robustness.

6.5 Resistance to Adversarial and Ambiguous Inputs

Future testing should examine how models respond to controlled changes in wording, punctuation, casing, paraphrase, negation, and ambiguous or adversarial language. Ji et al. (2021) identify adversarial training and data augmentation as relevant research directions. Systematic perturbation experiments could measure prediction stability rather than inferring robustness from aggregate benchmark performance.

6.6 Deployment Toolkits for Real-World Use

A substantial gap remains between notebook-based evaluation and responsible real-world deployment. Future work could explore APIs or moderation-support tools with explicit uncertainty handling, privacy protections, documented escalation policies, and human-in-the-loop review. Any such system would require prospective validation and domain-expert oversight before being used for intervention or clinical decision-making.

7 Conclusion

This paper presents a comparative evaluation of pretrained transformer architectures for classifying suicide-related Reddit content under a subreddit-derived proxy-labeling scheme. Using custom PyTorch fine-tuning and evaluation pipelines, I compared BERT, RoBERTa, and BERTweet against a TF-IDF + Logistic Regression baseline and examined their recorded performance, errors, and exploratory interpretability outputs.

In the recorded full-dataset comparison, RoBERTa achieved the highest suicide-class metrics, with precision of 0.996, recall of 0.992, and F1 of 0.994. BERT achieved 0.979 precision, 0.980 recall, and 0.980 F1, while BERTweet reached approximately 0.98 suicide-class F1. Using the common validation support, the recorded RoBERTa and BERT recall values correspond to approximately 178 and 457 false negatives respectively, although single-run experiments without variance estimates cannot establish the statistical robustness or cause of this difference. The classical TF-IDF + Logistic Regression baseline achieved 93% accuracy and 0.93 F1.

Qualitative review of selected BERT misclassifications identified examples involving informational discussion, general distress, and third-person framing. These examples provide hypotheses for more systematic error analysis rather than population-level conclusions about model behavior.

The interpretability experiments also revealed important limitations. The SHAP configuration used here produced near-zero token-level attributions in both high-probability and ambiguous examples and therefore did not provide sufficiently informative local explanations. The final-layer attention visualization showed interpretable visual structure, but attention weights alone do not establish causal explanations of a prediction.

Overall, the experiments demonstrate strong

predictive performance on this benchmark while also illustrating why benchmark performance must be separated from clinical validity, intention understanding, and trustworthy deployment. Future work should prioritize improved labeling, reproducible multi-seed evaluation, stronger interpretability methods, multilingual and temporal modeling, and human-in-the-loop validation.

Limitations

Several limitations constrain the conclusions of this study.

First, the dataset labels are derived from subreddit-based collection rather than clinical annotation. The task should therefore be interpreted as classification of a proxy label for suicide-related content, not diagnosis or direct measurement of individual suicidal ideation.

Second, dataset splitting and subset sampling used `random_state=42`, but model-training randomness was not comprehensively controlled through deterministic PyTorch and CUDA seeding. Each recorded configuration was evaluated only once, and no seed-variance or significance testing was performed.

Third, the full-dataset BERT and RoBERTa runs used a 256-token maximum sequence length while BERTweet used 128, so the models are not strictly sequence-length matched. Sequence length was not independently varied under otherwise identical conditions.

Fourth, the committed BERT/RoBERTa comparison notebook preserves the recorded model outputs and common data split but does not fully preserve tokenizer initialization state within its tokenization helper. This limits exact reconstruction of the execution environment from the committed source alone and is an additional reason to interpret the comparison as a recorded experimental result rather than a fully replicated benchmark.

Fifth, the 10K interpretability and misclassification analyses were performed in separate auxiliary BERT runs whose training configurations were not identical to every other subset experiment. Results from these analyses are therefore used for qualitative investigation rather than treated as interchangeable replications.

Sixth, the SHAP and attention analyses were applied only to BERT. The SHAP setup did not produce informative token-level explanations, and neither SHAP nor attention establishes causal un-

derstanding of model reasoning in these experiments.

Finally, the models were not evaluated prospectively, clinically, or in any real-world intervention setting. Their use here is exploratory and should not be interpreted as evidence of clinical validity or deployment readiness.

Appendix A: Example Misclassifications from Validation Set

The following posts were among the manually inspected examples misclassified by BERT on an auxiliary 2,000-post validation split from a balanced 10K sample. Excerpts are condensed with ellipses for length; wording within the retained excerpts is otherwise unaltered.

False Positives (predicted suicide, dataset label non-suicide)

“A possible suicide happened when I was going home from school... I’m pretty sure it was suicide... I’m posting here because teenagers are more likely to commit suicide... I simply want to raise awareness.”

Informational and awareness-oriented rather than an expression of the author’s own ideation, despite containing extensive suicide-related vocabulary.

“All my life I’ve wanted to travel around... They just announced that this year’s trip was cancelled. This sucks so much. We won’t have another chance to go. My dreams are shattered. Idk what to do.”

A post expressing disappointment that received a suicide prediction despite carrying the dataset’s non-suicide label.

“Sometimes when my day was extremely exhausting... there seem to be these voices/noises screaming literally so loud inside my head... It’s really awful and somewhat scary.”

A description of a distressing sensory experience without an explicit statement of suicidal ideation.

False Negatives (predicted non-suicide, dataset label suicide)

“Rick also stopt school because he was so devastated about his mom... He’s really depressed to a point that I don’t even know what to say anymore... I have to be really careful talking to my friend because he’s really depressed and I seriously don’t know if he might kill himself if I’m being too harsh.”

The suicide-related statement concerns another person and is embedded in a longer third-person narrative.

These manually inspected examples suggest that informational discussion and third-person framing are useful targets for future systematic error analysis. Because only a small number of errors were reviewed qualitatively, they should not be interpreted as estimates of how frequently these failure modes occur.

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